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SVAC SOLUTIONS, HYDERABAD
AI Engineer Appraisal – I Examinations, June – 2026
ARTIFICIAL INTELLIGENCE ENGINEERING
(Generative AI, LLMs & Model Context Protocol)

Time: 1 Hour

Max. Marks: 50

Note: This question paper contains two parts, A and B.
Part A is compulsory and carries 20 marks. Answer **all** questions in Part A.
Part B carries 30 marks. Answer **any FIVE** of the seven questions; each carries 6 marks.

PART – A

(10 × 2 = 20 Marks)

- 1.a) Define Large Language Models (LLMs) and outline the different approaches used to train them. [2]
- 1.b) Describe the procedure for computing attention outputs for a given sequence of vectors. [2]
- 1.c) Explain Context Engineering using the whiteboard analogy. [2]
- 1.d) Define LangChain. [2]
- 1.e) Briefly explain tokenization and its types. [2]
- 1.f) Define a vector database. [2]
- 1.g) List the transport types used in MCP communication. [2]
- 1.h) Define the following terms: (i) PBC (ii) Constitutional AI. [2]
- 1.i) Describe the anatomy of a SKILL.md file in Claude Code. [2]
- 1.j) Identify the various hook types and hook events available in Claude Code. [2]

PART – B

(5 × 6 = 30 Marks)

2. Distinguish between AI, ML, and Generative AI. [6]
3. Define a prompt. Discuss the various prompt engineering techniques and explain what constitutes the most effective prompt combination. [6]
4. Explain transformers and their architecture with a neat diagram. [6]
5. A simple transformer model is processing the sentence “Plan a candle light dinner for my partner.” Compute the attention for the word “dinner” using the vectors given below. Based on the resulting attention weights, interpret whether the model focuses more on “candle” or “plan.” [6]

$$Q = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad V = \begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$$

6. Differentiate between the RAG and Fine-Tuning mechanisms. [6]
7. Define MCP and its components. Explain the GitHub Server architecture with a neat diagram. [6]
8. Explain hooks with an example, and differentiate between Claude Code Skills and Claude Code Hooks. [6]